2 The Fourier transform

2.1 Definition and properties

Fourier series allow us to view periodic functions in a different light. In particular, they break up a function into sines and cosines (or complex exponentials), each of which has a frequency a multiple of $\Omega = 2\pi/T$, with the Fourier coefficients telling us how to weight each of these in the sum. In this sense, we can think of the sequence of Fourier coefficients as providing a different representation of the function, one in terms of the frequency instead of t. So what happens if we try to do this for non-periodic functions by taking $T \to \infty$?

Heuristically, by taking $T \to \infty$, the fundamental frequency Ω tends to zero and we must take into account all possible frequencies $\omega \in \mathbb{R}$. This is telling us that the analogue of the Fourier series for a non-periodic function must be an integral instead of a sum. This is the basis of the *Fourier transform* and the frequency representation of f, which we aim to motivate in this chapter.

First we want to generalise the idea of the Fourier coefficients, but now that we're dealing with functions that aren't periodic, we need to restrict to functions that are suitably well behaved over all of $\mathbb{R}$. This leads to the following definition.

Definition 2.1. The *Lebesgue space* $L^1(\mathbb{R})$ consists of all functions $f : \mathbb{R} \to \mathbb{C}$ such that

$$\int_{-\infty}^{\infty} |f(t)| \, \mathrm{d}t < \infty. \tag{2.1}$$

Suppose two functions f and g belong to $L^1(\mathbb{R})$. Then for any complex numbers α and β we have

$$\int_{-\infty}^{\infty} |\alpha f(t) + \beta g(t)| \, \mathrm{d}t \leq |\alpha| \int_{-\infty}^{\infty} |f(t)| \, \mathrm{d}t + |\beta| \int_{-\infty}^{\infty} |g(t)| \, \mathrm{d}t, \tag{2.2}$$

which is finite since $f, g \in L^1(\mathbb{R})$. Therefore, $L^1(\mathbb{R})$ is a vector space.

Definition 2.2. Let $f \in L^1(\mathbb{R})$. The *Fourier transform* of f is given by

$$\hat{f}(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t) \mathrm{e}^{-\mathrm{i}\omega t} \, \mathrm{d}t. \tag{2.3}$$

The Fourier transform generalises the Fourier coefficients in a natural way, extending the integration range to the whole real line and taking into account all possible frequencies ω. We can think of it as taking a function of time t and repackaging it into a function of frequency ω instead, so it is reasonable to say 'the Fourier transform lives in the frequency domain'. If it can be established that we can reconstruct $f(t)$ from $\hat{f}(\omega)$, then $\hat{f}(\omega)$ encodes all the relevant information about $f(t)$ in ways that might be more useful for certain problems. We will see this later, but for now let's compute an example.

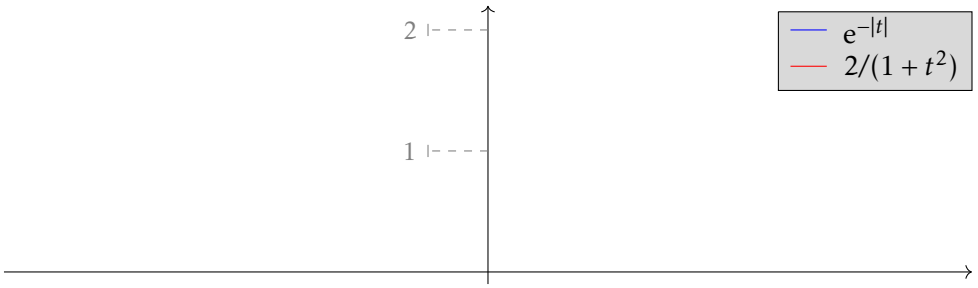


Figure 5: The function $f(t) = e^{-|t|}$ compared with its Fourier transform $\hat{f}(\omega) = 2/(1+\omega^2)$.

Example 2.1. Consider the function $f(t) = e^{-|t|}$. This belongs to $L^1(\mathbb{R})$ because

$$\int_{-\infty}^{\infty} |e^{-|t|}| \, dt = 2 \int_0^{\infty} e^{-t} \, dt = 2, \tag{2.4}$$

so we can calculate its Fourier transform. In particular, by splitting the integral range into $(-\infty, 0] \cup [0, \infty)$ we have

$$\begin{aligned} \hat{f}(\omega) = \int_{-\infty}^{\infty} e^{-|t| - i\omega t} \, dt &= \int_{-\infty}^{0} e^{(1-i\omega)t} \, dt + \int_0^{\infty} e^{-(1+i\omega)t} \, dt, \\ &= \left[\frac{e^{(1-i\omega)t}}{1-i\omega} \right]_{-\infty}^{0} + \left[-\frac{e^{-(1+i\omega)t}}{1+i\omega} \right]_0^{\infty}. \end{aligned} \tag{2.5}$$

Now note that $e^{(1-i\omega)t}$ vanishes as $t \to -\infty$ since the e^{-t} factor suppresses it. Likewise, $e^{-(1+i\omega)t}$ vanishes as $t \to \infty$, so we get

$$\hat{f}(\omega) = \frac{1}{1-i\omega} + \frac{1}{1+i\omega} = \frac{2}{1+\omega^2}. \tag{2.6}$$

Both f and $\hat{f}$ are plotted in fig. 5.

In this example, the resulting Fourier transform $\hat{f}(\omega) = 2/(1+\omega^2)$ also belongs to $L^1(\mathbb{R})$. This is not always the case! Consider the indicator function

$$\chi_{[-a,a]}(t) = \begin{cases} 1, & -a < t < a, \\ 0, & \text{otherwise.} \end{cases} \tag{2.7}$$

It is easy to show that the Fourier transform is given by

$$\hat{\chi}_{[-a,a]}(\omega) = \frac{2 \sin a\omega}{\omega}. \tag{2.8}$$

This function does not belong to $L^1(\mathbb{R})$ because the tails fall off as $1/|\omega|$ as $|\omega| \to \infty$. In general, the Fourier transform of an $L^1(\mathbb{R})$ function is a continuous function that vanishes at $\pm\infty$, due to the Riemann-Lebesgue lemma.

function	Fourier transform
$f(t)$	$\hat{f}(\omega)$
$\alpha f(t) + \beta g(t)$	$\alpha\hat{f}(\omega) + \beta\hat{g}(\omega)$
$f(at)$	$\frac{1}{\lvert a\rvert}\hat{f}\left(\frac{\omega}{a}\right)$
$f(t+b)$	$\mathrm{e}^{\mathrm{i}\omega b}\hat{f}(\omega)$
$tf(t)$	$\mathrm{i}\frac{\mathrm{d}\hat{f}}{\mathrm{d}\omega}$
$f'(t)$	$\mathrm{i}\omega\hat{f}(\omega)$
$\hat{f}(t)$	$2\pi f(-\omega)$
$(f * g)(t)$	$\hat{f}(\omega)\hat{g}(\omega)$

Table 1: Properties of the Fourier transform. Here α, β, a and b are constants with $a \neq 0$.

2.2 Transformation properties

Suppose we are given a function $f(t)$ and its Fourier transform $\hat{f}(\omega)$, but we want to find the Fourier transform of $g(t) = f(at+b)$ for $a \neq 0$. Instead of evaluating the integral directly, we can use the substitution rules for integrals to write $\hat{g}(\omega)$ in terms of $\hat{f}(\omega)$. Concretely, $\hat{g}(\omega)$ is given by

$$\hat{g}(\omega) = \mathcal{F}[f(at+b)] = \int_{-\infty}^{\infty} f(at+b)\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t. \tag{2.9}$$

Introducing the variable $u = at + b$, so $t = (u-b)/a$, we can rewrite the integral as

$$\hat{g}(\omega) = \int_{-\infty}^{\infty} f(u)\mathrm{e}^{-\mathrm{i}\omega(u-b)/a}\frac{\mathrm{d}u}{|a|} = \frac{\mathrm{e}^{\mathrm{i}\omega b/a}}{|a|}\hat{f}\left(\frac{\omega}{a}\right). \tag{2.10}$$

To justify why $|a|$ appears instead of a, note that if $a < 0$ the integration limits reverse to $(\infty, -\infty)$. However, we can switch the order of the limits by introducing a minus sign, taking the $1/a$ to $1/|a|$.

Now suppose we want to find the Fourier transform of $f'(t)$ instead. This can be done via integration by parts

$$\begin{aligned}\mathcal{F}[f'(t)] &= \int_{-\infty}^{\infty} f'(t)\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t, \\ &= \left[f(t)\mathrm{e}^{-\mathrm{i}\omega t}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \mathrm{i}\omega f(t)\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t.\end{aligned} \tag{2.11}$$

Since we've assumed $f(t) \in L^1(\mathbb{R})$, we must have that $f(t)$ vanishes at $\pm\infty$, so the boundary terms vanish. But now we're just left with the integral, which we recognise as

$$\mathcal{F}[f'(t)] = \mathrm{i}\omega\hat{f}(\omega). \tag{2.12}$$

This is perhaps the most important property of the Fourier transform: it turns derivatives by t into multiplication by $\mathrm{i}\omega$ in the frequency domain, turning highly difficult differential problems

into reasonably basic equations for $\hat{f}(\omega)$. This idea also holds in the opposite direction, taking the derivative of the Fourier transform formula gives

$$\frac{\mathrm{d}\hat{f}}{\mathrm{d}\omega} = \int_{-\infty}^{\infty} \frac{\mathrm{d}}{\mathrm{d}\omega}\left(f(t)\mathrm{e}^{-\mathrm{i}\omega t}\right)\mathrm{d}t = -\mathrm{i}\int_{-\infty}^{\infty} tf(t)\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t, \tag{2.13}$$

where we're allowed to move the derivative into the integral since $f \in L^1(\mathbb{R})$. The integral is now just the Fourier transform of $tf(t)$, so we have

$$\mathcal{F}[tf(t)] = \mathrm{i}\frac{\mathrm{d}\hat{f}}{\mathrm{d}\omega}, \tag{2.14}$$

thus turning multiplication by t into differentiation in the frequency domain.

Example 2.2. (i) We will find the Fourier transform of the Gaussian function $f(t) = \mathrm{e}^{-t^2/2}$. The integral expression for $\hat{f}(\omega)$ is given by

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} \mathrm{e}^{-t^2/2}\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t, \tag{2.15}$$

but this can be a little tricky to evaluate. Instead, we can indirectly find $\hat{f}$ by noting that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\mathrm{e}^{-t^2/2}\right) = -t\mathrm{e}^{-t^2/2} = -tf(t) \tag{2.16}$$

Using the rules for Fourier transforms of derivatives and $tf(t)$ we obtain the differential equation

$$\frac{\mathrm{d}\hat{f}}{\mathrm{d}\omega} = -\omega\hat{f}(\omega). \tag{2.17}$$

Rearranging this, we get $\hat{f}'(\omega)/\hat{f}(\omega) = -\omega$ and recognise the left-hand side as the derivative of $\log\hat{f}(\omega)$. Integrating this rearrangement, we arrive at

$$\log\hat{f}(\omega) = -\frac{\omega^2}{2} + \log C \Longrightarrow \hat{f}(\omega) = C\mathrm{e}^{-\omega^2/2}, \tag{2.18}$$

where $\log C$ is the constant of integration. We've therefore determined, up to a constant, that the Fourier transform of a Gaussian is another Gaussian. By using the fact that $\hat{f}(0) = \int_{-\infty}^{\infty}\mathrm{e}^{-t^2/2}\,\mathrm{d}t = \sqrt{2\pi}$, we see that $C = \sqrt{2\pi}$. In summary, we have

$$\mathcal{F}[\mathrm{e}^{-t^2/2}] = \sqrt{2\pi}\mathrm{e}^{-\omega^2/2}. \tag{2.19}$$

(ii) Now consider the functions $g(t) = \mathrm{e}^{-t^2}$ and $h(t) = \mathrm{e}^{-(t^2+2t)/2}$. Both of these functions can be written in terms of $f(t)$ as

$$g(t) = f(t\sqrt{2}), \qquad h(t) = \mathrm{e}^{-((t+1)^2-1)/2} = \mathrm{e}^{1/2}f(t+1), \tag{2.20}$$

which the transformation rules of Fourier transforms imply

$$\hat{g}(\omega) = \sqrt{\pi}\mathrm{e}^{-\omega^2/4}, \quad \hat{h}(\omega) = \sqrt{2\pi}\mathrm{e}^{-\omega^2/2+\mathrm{i}\omega+1/2} \tag{2.21}$$

2.3 Convolution theorem

Given that the Fourier transform turns differentiation into multiplication by $\mathrm{i}\omega$, it is natural to ask which other operations it simplifies. Here we will examine its interaction with the *convolution* of two functions.

Definition 2.3. The *convolution* of two functions $f, g : \mathbb{R} \to \mathbb{C}$ is

$$(f * g)(t) \doteq \int_{-\infty}^{\infty} f(u)g(t-u)\,\mathrm{d}u. \tag{2.22}$$

Admittedly, the convolution is not immediately straightforward to understand, but if we break the definition down a bit we can get some intuition.

First of all, we can plot f as a function of u since there is no t-dependence. Then we can plot $g(-u)$ on top of this, which would correspond to $t = 0$ in the definition. The $t = 0$ convolution is then the area under the curve $f(u)g(-u)$, so the more $g(-u)$ and $f(u)$ overlap, the larger the convolution is. Finally, we can introduce the t-dependence by sliding the graph of $g(-u)$ across $f(u)$ to calculate the area under $f(u)g(t-u)$, calculating the overlap for different translations of $g(-u)$. The following example illustrates this in a simple case.

Example 2.3. Consider the function $f(t) = \chi_{[-1,1]}(t)$. The self-convolution $f * f$ is given by

$$(f * f)(t) = \int_{-\infty}^{\infty} \chi_{[-1,1]}(u)\chi_{[-1,1]}(t-u)\,\mathrm{d}u. \tag{2.23}$$

The first term in the integrand is non-zero when $-1 \leq u \leq 1$, while the second term is non-zero when $-1 \leq t-u \leq 1$, which we can rewrite as $-1+t \leq u \leq 1+t$. Since both of these need to be satisfied for the integrand to be non-zero, we must look at when these two ranges overlap. Firstly, if $|t| > 2$, then the two ranges do not overlap and $(f * f)(t) = 0$. Now take $-2 < t < 2$, then the overlap is

$$[-1,1] \cap [-1+t, 1+t] = [\max\{-1, -1+t\}, \min\{1, 1+t\}] \doteq [u_1, u_2]. \tag{2.24}$$

That is, if $t < 0$ then the intersection is $[-1, 1+t]$, while $t > 0$ gives $[-1+t, 1]$. Therefore, for $-2 < t < 2$ the convolution is

$$(f * f)(t) = \int_{u_1}^{u_2} \mathrm{d}u = \min\{1, 1+t\} - \max\{-1, -1+t\} = \begin{cases} 2-t, & t \geq 0 \\ 2+t & t < 0. \end{cases} \tag{2.25}$$

This is just $2 - |t|$, so we can write the convolution as

$$(f * f)(t) = (2 - |t|)\chi_{[-2,2]}(t), \tag{2.26}$$

where we've included $\chi_{[-2,2]}(t)$ to ensure it is zero outside of $[-2,2]$.

Now we turn to how the convolution behaves under Fourier transformation. First note that if $f, g \in L^1(\mathbb{R})$, we can show that $f * g$ also belongs to $L^1(\mathbb{R})$. Concretely, we have

$$\int_{-\infty}^{\infty} |(f*g)(t)|\,\mathrm{d}t = \int_{-\infty}^{\infty} \left| \int_{-\infty}^{\infty} f(u)g(t-u)\,\mathrm{d}u \right| \mathrm{d}t \leq \int_{\mathbb{R}^2} |f(u)||g(t-u)|\,\mathrm{d}u\,\mathrm{d}t. \tag{2.27}$$

We can eliminate t by changing variables to $s = t - u$, yielding

$$\int_{-\infty}^{\infty} |(f*g)(t)|\,\mathrm{d}t \leq \left(\int_{-\infty}^{\infty} |f(u)|\,\mathrm{d}u \right) \left(\int_{-\infty}^{\infty} |g(s)|\,\mathrm{d}s \right), \tag{2.28}$$

which is finite by assumption. Therefore, $f * g \in L^1(\mathbb{R})$ and we can attempt to Fourier transform it. In doing so, we arrive at

$$\begin{aligned} \widehat{(f*g)}(\omega) &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(u)g(t-u)\,\mathrm{d}u \right) \mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t, \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u)g(s)\mathrm{e}^{-\mathrm{i}\omega(s+u)}\,\mathrm{d}u\,\mathrm{d}s, \end{aligned} \tag{2.29}$$

by making the change of variables $s = t - u$. This integral factorises nicely as

$$\begin{aligned} \widehat{(f*g)}(\omega) &= \left(\int_{-\infty}^{\infty} f(u)\mathrm{e}^{-\mathrm{i}\omega u}\,\mathrm{d}u \right) \left(\int_{-\infty}^{\infty} g(s)\mathrm{e}^{-\mathrm{i}\omega s}\,\mathrm{d}s \right), \\ &= \hat{f}(\omega)\hat{g}(\omega). \end{aligned} \tag{2.30}$$

As such, convolution of functions gets mapped to products of Fourier transforms. For our purposes, this is the defining feature of the convolution: when we see a product of Fourier transforms, we should immediately think of the convolution.

Example 2.4. Using the previous example, we have

$$\mathcal{F}\left[(2-|t|)\chi_{[-2,2]}(t)\right] = \mathcal{F}[\chi_{[-1,1]}(t)]^2 = \frac{4\sin^2\omega}{\omega^2}. \tag{2.31}$$

2.4 Fourier inversion theorem

The Fourier transform is the natural analogue of the Fourier coefficients for periodic functions, but in that case we were also able to use these coefficients to reconstruct f via the Fourier series. We can also do this with the Fourier transform.

First we will sketch the idea and then state the theorem more concretely. Suppose we have an $L^1(\mathbb{R})$ function $f : \mathbb{R} \to \mathbb{C}$ that we wish to write down a Fourier series for. We can do this by defining f_T to be f restricted to the range $[-T/2, T/2]$ and extended periodically. Then, assuming the convergence of the Fourier series, we have

$$f_T(t) = \frac{1}{2\pi} \sum_{n\in\mathbb{Z}} \Omega \mathrm{e}^{\mathrm{i}n\Omega t} g(n\Omega), \qquad g(\omega) = \int_{-T/2}^{T/2} f(\tau)\mathrm{e}^{-\mathrm{i}\omega\tau}\,\mathrm{d}\tau, \tag{2.32}$$

where we've written the usual $1/T$ factor as $\Omega/2\pi$. Recall that the Riemann integral can be obtained by sampling a function at a number of equally distanced points, multiplying these samples by the gap between points and then taking the limit to close the gap. This is essentially what is going on here: the gap between points is given by Ω and we are sampling $e^{i\omega t}g(\omega)$ at $\omega = n\Omega$. The only caveat is that $g(\omega)$ changes as T increases due to the integration range being $[-T/2, T/2]$. We can get around this for T suitably large, by writing $g(\omega)$ as

$$g(\omega) = \hat{f}(\omega) - \int_{-\infty}^{-T/2} f(\tau)e^{-i\omega\tau}\,d\tau - \int_{T/2}^{\infty} f(\tau)e^{-i\omega\tau}\,d\tau, \tag{2.33}$$

and noting that these two corrections to $\hat{f}(\omega)$ must vanish as $T \to \infty$ since $f \in L^1(\mathbb{R})$. This leaves us with $g(\omega) \to \hat{f}(\omega)$ as $T \to \infty$ and we obtain

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\omega)e^{i\omega t}\,d\omega. \tag{2.34}$$

This is our analogue of the Fourier series, only now it is an integral that sums over every possible frequency ω. In doing this, we've been quite sketchy with the details, but the rigorous result is as follows.

Theorem 2.1. Let $f : \mathbb{R} \to \mathbb{C}$ be a continuous absolutely integrable function such that $\hat{f}$ is also continuous and absolutely integrable. Then

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\omega)e^{i\omega t}\,d\omega. \tag{2.35}$$

This is known as the *Fourier inversion theorem* and allows us to reconstruct $f(t)$ from its Fourier transform. There are various ways to prove this rigorously, but they are outside the scope of this course.

The fact that the inverse Fourier transform looks like the usual Fourier transform with $\omega \mapsto -\omega$ has an important consequence. Given a function's Fourier transform $\hat{f}(\omega)$, we can try to Fourier transform this as usual

$$\mathcal{F}[\hat{f}] = \int_{-\infty}^{\infty} \hat{f}(\omega)e^{-i\omega t}\,d\omega. \tag{2.36}$$

But as we've just remarked, this is just the inverse transform up to a minus sign in the exponent and a factor of 2π. We therefore get

$$\mathcal{F}^2[f] = \mathcal{F}[\hat{f}] = 2\pi f(-t). \tag{2.37}$$

In other words, if we recognise a function $g(t)$ as the Fourier transform of a function $f(t)$, then $\hat{g}(\omega) = 2\pi f(-\omega)$. This is sometimes referred to as the ***duality property*** of Fourier transforms.

2.5 Plancherel's theorem

Parseval's theorem also has a natural extension to Fourier transforms, only now it relates two integrals rather than an integral and a sum. To state this, we need to return to L^2 functions.

Definition 2.4. The space $L^2(\mathbb{R})$ consists of all functions $f : \mathbb{R} \to \mathbb{C}$ such that

$$\int_{-\infty}^{\infty} |f(t)|^2 \, \mathrm{d}t < \infty. \tag{2.38}$$

Like before, $L^2(\mathbb{R})$ is an inner product space with the inner product

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(t)\overline{g(t)} \, \mathrm{d}t, \tag{2.39}$$

satisfying the Cauchy-Schwarz inequality $|\langle f, g \rangle| \leq \sqrt{\langle f, f \rangle \langle g, g \rangle}$. Crucially, $L^2(\mathbb{R})$ is not a subspace of $L^1(\mathbb{R})$, as the following example illustrates.

Example 2.5. While many functions belong to both $L^1(\mathbb{R})$ and $L^2(\mathbb{R})$, we will construct two examples that only belong to one of the two spaces. Consider the two functions

$$f_1(x) = \begin{cases} \frac{1}{\sqrt{x}}, & 0 < x \leq 1, \\ 0, & \text{otherwise.} \end{cases} \qquad f_2(x) = \begin{cases} \frac{1}{x}, & x \geq 1, \\ 0, & \text{otherwise.} \end{cases} \tag{2.40}$$

The L^1-norm of f_1 is given by $\int_0^1 1/\sqrt{x} \, \mathrm{d}x = 2$, while the square integral is

$$\int_{-\infty}^{\infty} |f_1(x)|^2 \, \mathrm{d}x = \int_0^1 \frac{1}{x} \, \mathrm{d}x = [\log x]_0^1, \tag{2.41}$$

which is infinite. Therefore, f_1 belongs to $L^1(\mathbb{R})$, but not $L^2(\mathbb{R})$. For f_2 it is the opposite: it belongs to $L^2(\mathbb{R})$ but not $L^1(\mathbb{R})$.

Theorem 2.2. Let $f, g \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. Then

$$\int_{-\infty}^{\infty} f(t)\overline{g(t)} \, \mathrm{d}t = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega)\overline{\hat{g}(\omega)} \, \mathrm{d}\omega \tag{2.42}$$

and

$$\int_{-\infty}^{\infty} |f(t)|^2 \, \mathrm{d}t = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{f}(\omega)|^2 \, \mathrm{d}\omega. \tag{2.43}$$

Proof. We will assume that f and g are both continuous so the Fourier inversion theorem holds. In this case, we can write

$$\overline{g(t)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \overline{\hat{g}(\omega)} \mathrm{e}^{-\mathrm{i}\omega t} \, \mathrm{d}\omega, \tag{2.44}$$

by moving the complex conjugation into the integrand. Substituting this into the left-hand side

of eq. (2.42) gives

$$\begin{aligned}\int_{-\infty}^{\infty} f(t)\overline{g(t)}\,\mathrm{d}t &= \int_{-\infty}^{\infty} f(t)\left(\frac{1}{2\pi}\int_{-\infty}^{\infty}\overline{\hat{g}(\omega)}\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}\omega\right)\mathrm{d}t,\\ &= \int_{-\infty}^{\infty}\left(\frac{1}{2\pi}\int_{-\infty}^{\infty} f(t)\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t\right)\overline{\hat{g}(\omega)}\,\mathrm{d}\omega.\end{aligned} \tag{2.45}$$

Here we're allowed to exchange the order of integration since both $f(t)$ and $\hat{g}(\omega)$ are assumed to be in $L^1(\mathbb{R})$. We now recognise the integral over t as the Fourier transform of f, therefore

$$\int_{-\infty}^{\infty} f(t)\overline{g(t)}\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty}\hat{f}(\omega)\overline{\hat{g}(\omega)}\,\mathrm{d}\omega. \tag{2.46}$$

Taking $g = f$ then gives the second result. □

Plancherel's theorem states that the Fourier transform preserves the $L^2(\mathbb{R})$ inner product up to a factor of 2π. Consequently, the Fourier transform of an $L^2(\mathbb{R})$ function also belongs to $L^2(\mathbb{R})$. Contrast this with the $L^1(\mathbb{R})$ case where the Fourier transform is often not absolutely integrable, with the Fourier transform of $\chi_{[-a,a]}(t)$ being a simple example.

Example 2.6. (i) We saw that $\chi_{[-a,a]}(t)$ has Fourier transform $2\sin a\omega/\omega$ earlier. Taking $f(t) = \chi_{[-a,a]}(t)$ and $g(t) = \chi_{[-b,b]}(t)$, Plancherel's formula states

$$\int_{-\infty}^{\infty}\chi_{[-a,a]}(t)\chi_{[-b,b]}(t)\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{4\sin a\omega\sin b\omega}{\omega^2}\,\mathrm{d}\omega. \tag{2.47}$$

The right-hand side is difficult to compute, but the left-hand side is easy. The integrand is non-zero only when $|t| < \min\{a,b\}$, so

$$\int_{-\infty}^{\infty}\frac{\sin a\omega\sin b\omega}{\omega^2}\,\mathrm{d}\omega = \pi\min\{a,b\}. \tag{2.48}$$

(ii) The function $f(t) = \mathrm{e}^{-|t|}$ has Fourier transform $\hat{f}(\omega) = 2/(1+\omega^2)$. Integrating $1/(1+\omega^2)^2$ over $\mathbb{R}$ is doable via substitution or integration by parts, but using Plancherel's theorem it is even easier. Namely, we have

$$\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega}{(1+\omega^2)^2} = \frac{\pi}{2}\int_{-\infty}^{\infty}\mathrm{e}^{-2|t|}\,\mathrm{d}t = \pi\int_0^{\infty}\mathrm{e}^{-2t}\,\mathrm{d}t = \frac{\pi}{2}. \tag{2.49}$$

If you're brave, you can do the integral directly to confirm this matches.